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How does a bike-share ride to rapid success?

Cyclistic is the Chicago bike-share company with more than 5,800 bikes at close to 700 stations. The company has three different pricing schemes: single-ride passes, full-day passes and annual memberships. Financial analysts have determined that annual members are much more profitable than casual riders. Therefore, the marketing team is looking to develop a strategy to convert casual riders into annual members to support the company's future growth.

To accomplish this goal, the marketing analytics team should know the difference in the use of Cyclistic bikes by casual riders and annual members. The analysis will identify usage patterns and behavioral differences between the two customer segments, enabling the company to develop data-driven marketing strategies to boost annual membership adoption, improve client retention, and optimize company profitability.

The data from the historical trip data for Cyclistic will be used for the analysis. The analysis will consider variables such as ride duration, ride frequency, day of the week, time of day, bike type, and station usage to find meaningful trends that will support business recommendations.

Key Stakeholders:

- Lily Moreno (Director of Marketing): Responsible for developing marketing campaigns and increasing annual memberships.
- Cyclistic Marketing Analytics Team: Responsible for collecting, analyzing, and reporting data to support business decisions.
- Cyclistic Executive Team: Responsible for evaluating and approving the proposed marketing strategy.

The data used for this analysis was obtained from the Divvy Trip Data repository which is public, the data is considered reliable because it comes from the official bike-sharing system and it is original since it is collected directly from the service, The dataset contains one year of trip records and for analysis I selected the most recent dataset that was of 2025.

The data is in monthly csv format where each row represents a single bike trip. The dataset provides the following information: ride ID, rideable type, start and end timestamps, start and end station names and IDs, station coordinates, and rider type. These variables allow for the comparison between casual riders and annual members to identify behavioral patterns that can support marketing strategies to increase annual memberships.

These datasets are publicly available under a public data license and have been anonymized meaning they do not contain personally identifiable information protecting user privacy while allowing meaningful analysis.

To check the integrity of the data I did an initial exploration to verify that all monthly datasets shared the same structure and variables. Initial exploration also showed presence of some missing values which will be addressed during the data cleaning phase before the analysis starts.

Due to the dataset containing millions of records in twelve monthly tables, SQL (BigQuery) was chosen for the cleaning and analyzing process. SQL is an efficient way to query, transform and aggregate data, and can handle massive datasets much faster than spreadsheet software.

First, I imported and joined the 12 months datasets into one table using UNION ALL. An initial exploration of all the twelve datasets was carried out to check the integrity of the data to ensure that they had the same structure and variables.

Missing values were identified in the following variables:

- start_station_name: 1,184,673
- end_station_name: 1,243,305
- start_station_id: 1,184,673
- end_station_id: 1,243,305

The variables were kept because the missing values are only in the station information and these variables do not affect the main objective of the analysis, which is to compare the riding behavior of casual riders and annual members. Imputation of values may lead to a biased dataset and deletion of missing values may reduce the dataset size significantly.

Ride IDs were checked for duplicates but no duplicates were found.

There were 29 rides where the ended_at timestamp occurred before started_at. So after investigation all of these records occur on the same day which was November 2, 2025 this coinciding with the daylight saving time transition in Chicago . These records were kept as timestamp anomalies due to the clock change. They were flagged with an invalid_duration indicator so that they do not affect the calculations of ride durations.

There were also 12 trips having durations greater than 24 hours. Upon investigation these trips ranged between 25 and 26 hours. There was no clear evidence of data corruption, and the number of records affected was miniscule compared to the entire dataset, so these rides were kept and flagged for further analysis.

To facilitate the analysis, new variables were created from the original timestamp fields:

Variable	Description
ride_date	Date when the ride started
ride_length_minutes	Ride duration in minutes
day_of_week	Day of the week when the rides started
month	Month in which the ride started
hour	Hour when the ride started
day_type	Classifies rides as Weekday or Weekend

Variable	Description
Invalid_duration	Flags rides affected by timestamp inconsistencies
long_ride_over_24h	Flags rides longer than 24 hours
complete_station_data	Indicates whether station information is complete

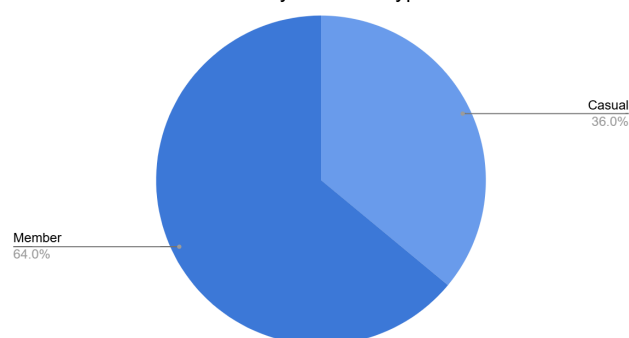
The resulting processed dataset retained all original records, and recorded data quality problems using indicator variables rather than deleting observations. This ensured the integrity of the dataset and only relevant records would be excluded from specific analyses when appropriate.

FINAL DATASET SUMMARY	
Metric	Results
Total records	5,552,994
Duplicates rides	0
Missing values	4,855,956
Invalid timestamps	29 (flagged)
Long rides (>24h)	12 (flagged)
New variables created	9

After cleaning and transforming the data, I did an exploratory data analysis (EDA) in SQL and Google Sheets visualizations. The goal was to see if there were any behavioral differences between casual riders and annual members in terms of ride frequency, duration, bike type preference, time of usage, and popular station locations.

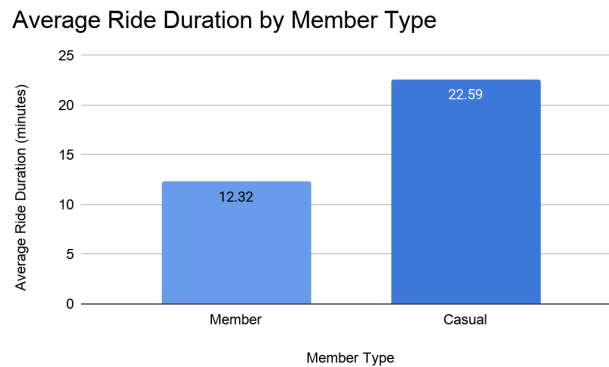
Now for the analysis first I checked the total rides of each type member giving as result:

Distribution of Total Rides by Member Type



Annual members accounted for 64% of all trips, taking nearly twice as many rides as casual riders in 2025. This suggests that the casual riders used the bike-sharing system less often than the annual members, who take advantage of Cyclistic to incorporate it into their daily transportation needs.

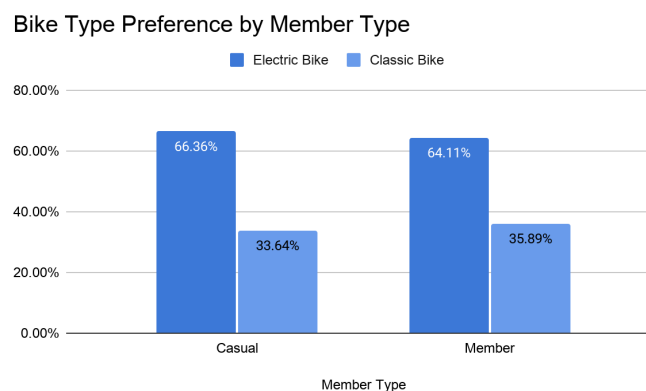
Then I analyze the average duration time of a ride for each member type and I get:



Casual riders spend nearly twice as long per trip than annual members do. This suggests that casual riders are using Cyclistic bikes for leisure, recreational or occasional trips, whereas annual members are more likely to use the service for short, routine trips.

This behavioural difference points to marketing strategies targeting the value and cost savings of an annual membership for casual riders who regularly do longer recreational trips. Highlight the value of unlimited rides to turn these users into annual members.

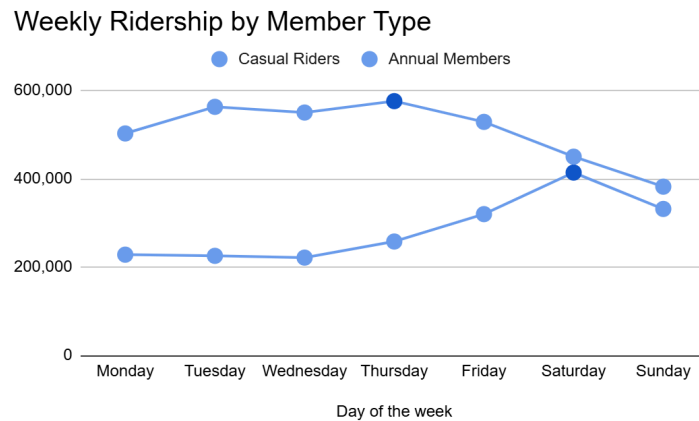
Then I analyze the Bike Type preference of each member type:



Both customer groups have very similar preferences for electric bikes. Two thirds of all rides are made with electric bikes and one third with classic bikes.

The preferences of both groups are very similar and bike type does not seem to be a major differentiator between casual riders and annual members.

This was followed by an analysis of weekly ridership by each member type:

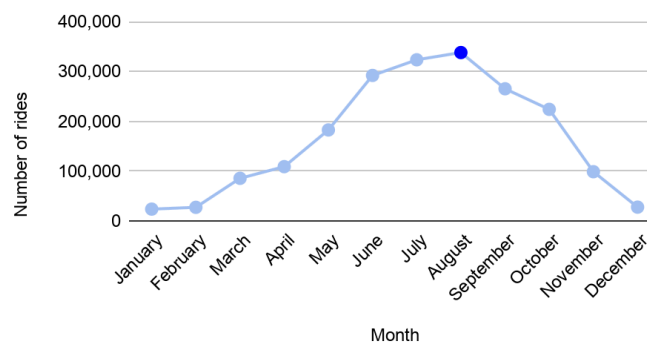


The line graph indicates clear differences in riding behavior in the week. Annual members ride more on weekdays than casual riders and the highest point is on Thursday, then it declines over the weekend. By contrast, casual riders exhibit relatively steady ridership from Monday to Thursday, a marked increase on Friday and a peak on Saturday.

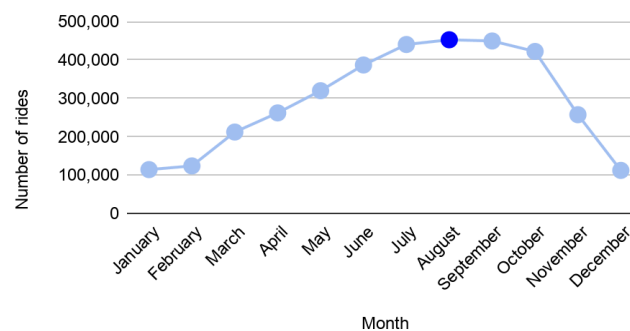
These trends suggest that members tend to use Cyclistic bikes for daily commutes and transportation needs, while casual riders are more likely to use bikes for recreational and leisure activities on the weekends.

I then checked the most popular month for each member type to use the bikes and the results were:

Monthly Ridership - Casual Riders



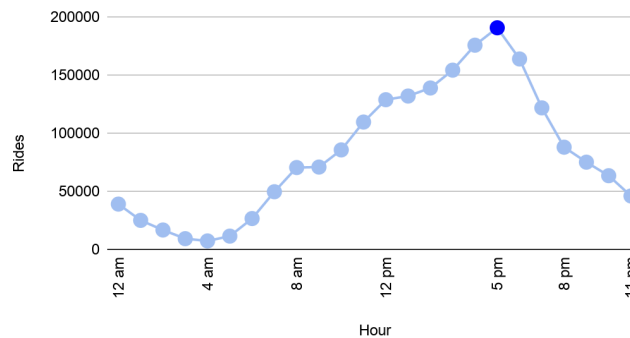
Monthly Ridership - Annual Members



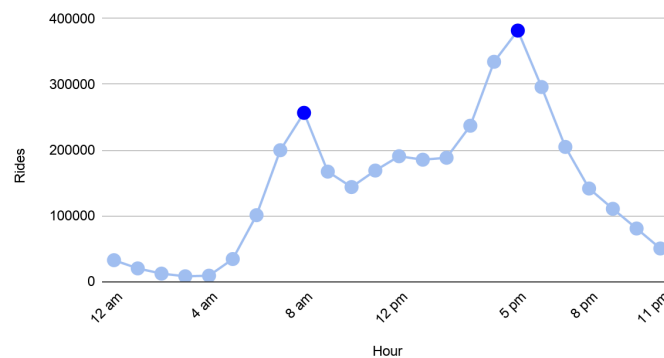
Both customer groups show a distinct seasonal riding pattern throughout the year. January and December have the fewest rides, while ridership increases gradually from March, peaking in August. This is a seasonal trend, meaning that all types of memberships use bikes more in the warmer months and less in the colder months.

Then I look at rides per hour:

Number of rides per Hour - Casual Riders



Number of rides per Hour - Annual Members



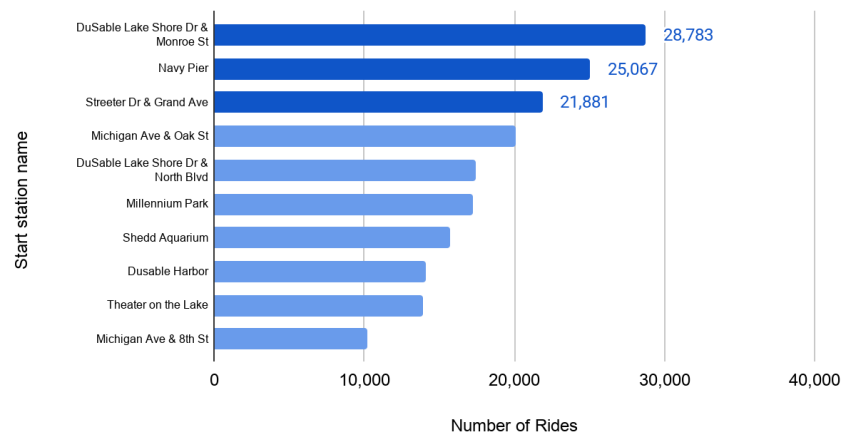
There are distinct hourly riding patterns for annual members and casual riders, as shown by the line graphs. Annual members use bikes in a bimodal pattern with peaks around 8:00 AM and even more so around 5:00 PM. This suggests that annual members are primarily using Cyclistic bikes for daily transportation as these peaks occur during typical commuting hours.

Unlike casual riders have a steady increase of ridership throughout the day with the highest amount of rides occurring around 5:00 PM. By contrast, casual riders do not show a strong morning peak, suggesting that they are more likely to use the bikes later in the day for recreational or non-routine trips.

These results are consistent with the previous reviews of weekdays and ride length, supporting the conclusion that annual members primarily use the Cyclistic bikes for transportation purposes, while casual riders primarily use the bikes for recreational and occasional trips.

And finally just to make a clear recommendation for the advertising I made the top 10 more popular stations and the results were:

Top 10 Most Popular Start Stations for Casual Riders



Casual riders usually begin in the vicinity of Chicago's waterfront and popular tourist destinations, like DuSable Lake Shore Drive, Navy Pier, Millennium Park and Shedd Aquarium. This suggests that the Cyclistic bikes are primarily enjoyed by recreational riders, not the daily commuters. These places are therefore good candidates for marketing campaigns aimed at annual memberships during periods of heavy tourist traffic.

Analysis summary

The analysis revealed several important behavioral differences between casual riders and annual members:

- Annual members complete significantly more rides throughout the year.
- Casual riders spend more time per ride.
- Both groups strongly prefer electric bikes.
- Annual members mainly ride during weekdays and commuting hours.
- Casual riders ride more frequently on weekends and during the afternoon.
- Ridership increases considerably during summer months.
- Casual riders frequently begin their trips at tourist attractions and waterfront locations.

These findings provide valuable insights that can be used to design targeted marketing strategies focused on converting casual riders into annual members.

The analysis shows that annual members and casual riders use Cyclistic bikes for different reasons. The majority of annual members use the service mainly for daily commutes and ride more frequently on weekdays and during commuting hours. In comparison, leisure and recreational activities are the main reasons for casual riders to use the bikes, particularly during weekends, afternoons, and summer months. These behavioral differences provide a niche for targeted marketing strategies to convert casual riders into annual members.

Recommendations:

- Develop summer and weekend seasonal marketing campaigns to capitalize on the peak of casual ridership. Promotional offers emphasizing the cost savings of an annual membership can induce frequent seasonal riders to subscribe.
- Put ads and membership promotions at the most popular stations for casual riders, like Navy Pier, Millennium Park, DuSable Lake Shore Drive and Shedd Aquarium. These are places where there are lots of casual riders, so they are great places to promote annual memberships.
- Casual riders take almost twice as long per trip as annual members. Cyclistic should stress that annual memberships offer unlimited rides and represent better value for money over time for users who enjoy long recreational rides regularly.

Further data relating to customer age, trip purpose, weather, tourist seasons, promotion campaign history and customer surveys would give better insights into rider behaviour and help to develop more personalized marketing strategies.

Overall, the analysis implies that casual riders use Cyclistic for recreation, whereas annual members use it for daily transportation. These behavioral differences provide clear opportunities for targeted marketing campaigns to nudge casual riders toward annual membership.